8f. Type Stability with Tuples

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INTRODUCTION

A function is considered type stable when, given the types of its arguments, the compiler can accurately predict single concrete types for its expressions. This definition, while universal, takes on different forms when applied to specific objects. So far, we've exclusively dealt with scalars and vectors, whose conditions for type stability are relatively straightforward.

In this section, we begin our analysis of type stability for other data structures. In particular, we consider tuples, whose coverage automatically encompasses **named tuples**. Guaranteeing type stability with tuples is more nuanced compared to vectors, as their type characterization demands more information. Its exploration will challenge our understanding of type stability, demanding a clear grasp of its definition and subtleties.

> **Warning!** - Tuples Are Only Suitable For Small Collections
> Remember that tuples should only be used for collections that comprise a few elements. Using them for large collections will result in significant performance degradation or directly trigger fatal errors.

COMPARING TUPLES AND VECTORS

Tuples and vectors are the most common forms of collections in Julia. While both fulfill a similar purpose, they differ significantly in their underlying implementation. In particular, tuples tend to outperform vectors when working with small objects, by avoiding the memory-allocation overhead incurred by vectors. This advantage will be explained in more depth when discussing static vectors, which are essentially tuples that can be manipulated as vectors.

Another key distinction is that tuples possess a more intricate type system in comparison to vectors. To see this, let's compare the information needed to describe each type.

Vectors represent collections of elements sharing a *homogeneous* type and exhibiting a variable size. Thus, the information needed to describe the types of vectors is relatively minor. For instance, a type like `Vector{Float64}` establishes that *all* elements must have type `Float64`, without any restriction on the number of elements to be contained.

For their part, tuples are fixed-size collections that can accommodate *heterogeneous* types. This makes the characterization of a tuple's type more demanding, requiring both the number of elements and the type of *each* element. For instance, the variable `tup = ("hello", 1)` has type `Tuple{String,`

`Int64}`, indicating that the first element has type `String` and the second one `Int64`. Furthermore, it implicitly sets the number of elements to two, as there's no possibility of appending or removing elements.

The fact that the number of elements is part of the type becomes clear when tuples contain `N` elements of the same type `T`. For this case, Julia provides the convenient alias `Ntuple{N, Float64}`, which is just syntactic sugar for `Tuple{T, T,...,T}` where `T` appears `N` times. [1]

In the following, we show that the choice between tuples and vectors may have different implications for type stability.

TUPLE SLICES WITH MIXED TYPES CAN STILL BE TYPE STABLE

One key difference between tuples and vectors in Julia lies in how they handle type information. While tuples explicitly define the type of each individual element, vectors require all elements to be of a uniform type.

Because vectors must maintain a consistent type throughout, attempting to store mixed concrete types within a single vector compels Julia to determine a common type that accommodates them all. For example, if you create a vector containing both `Int64` and `Float64`, Julia will infer the type of the vector as `Vector{Float64}`, the most general type encompassing both integer and float types.

However, when dealing with highly diverse element types within a vector, this process can lead to less efficient behavior. In extreme cases, Julia might resort to using the abstract type `Any`, resulting in a `Vector{Any}`. Working with vectors like this is extremely undesirable from a performance point of view.

This issue particularly affects vector slices, as they inherit the type information from their parent vector. Thus, if the parent vector has been widened to a more general type like `Vector{Any}`, operations performed on those slices will also be subject to that same type instability. The behavior contrasts sharply with **slices of tuples, where each element within the slice retains its concrete type.**

TUPLE

```julia
tup    = (1, 2, "hello")        # type is `Tuple{Int64, Int64, String}`

foo(x) = sum(x[1:2])

@code_warntype foo(tup)         # type stable (output is `Int64`)
```

VECTOR

```julia
vector = [1, 2, "hello"]        # type is `Vector{Any}`

foo(x) = sum(x[1:2])

@code_warntype foo(vector)      # type UNSTABLE
```

TUPLES CONTAIN MORE INFORMATION THAN VECTORS

Given the differences in type information, conversions between tuples and vectors can pose several challenges for type stability.

To see this, let's start with the simplest case, where a tuple is converted into a vector. The outcome of this conversion is predictable, stemming directly from our previous analysis: type stability will be preserved when the tuple contains all elements having the same type or when heterogeneous types can be promoted to a common concrete type.

For the examples, recall that **each type automatically generates a function** that transforms variables into the corresponding type. For instance, the function `Vector` converts variables to this type.

TYPE-HOMOGENEOUS TUPLES

```
tup = (1, 2, 3)                # `Tuple{Int64, Int64, Int64}` or just `NTuple{3, Int64}`


function foo(tup)
    x = Vector(tup)            # 'x' has type `Vector(Int64)}`
    sum(x)
end

@code_warntype foo(tup)        # type stable
```

TYPE PROMOTION

```
tup = (1, 2, 3.5)              # `Tuple{Int64, Int64, Float64}`


function foo(tup)
    x = Vector(tup)            # 'x' has type `Vector(Float64)}`
    sum(x)
end

@code_warntype foo(tup)        # type stable
```

TYPE-HETEROGENEOUS TUPLES

```
tup = (1, 2, "hello")          # `Tuple{Int64, Int64, String}`


function foo(tup)
    x = Vector(tup)            # 'x' has type `Vector(Any)}`
    sum(x)
end

@code_warntype foo(tup)        # type UNSTABLE
```

Likewise, **creating a tuple from a vector will inevitably cause type instability**, regardless of the vector's characteristics. The reason is that vectors don't store information about the number of elements they contain. Consequently, when attempting to construct a tuple from a vector, the compiler must account for the possibility of varying numbers of arguments. The result is that each potential number of elements corresponds to a distinct concrete type for the tuple.

VECTOR WITH NON-PRIMITIVE TYPES

```
x   = [1, 2, "hello"]           # 'Vector{Any}' has no info on each individual type


function foo(x)
    tup = Tuple(x)              # 'tup' has type `Tuple`

    sum(tup[1:2])
end

@code_warntype foo(x)           # type UNSTABLE
```

VECTOR WITH PRIMITIVE TYPES

```
x   = [1, 2, 3]                 # 'Vector{Int64}' has no info on the number of elements


function foo(x)
     tup = Tuple(x)                # 'tup' has type `Tuple{Vararg(Int64)}` (`Vararg` means
"variable arguments")

    sum(tup[1:2])
end

@code_warntype foo(x)           # type UNSTABLE
```

ADDRESSING VARIABLE ARGUMENTS: DISPATCH BY VALUE

A key takeaway from the previous subsection is that defining tuples from vectors invariably introduces type instability. A simple remedy for this is to convert tuples outside the function, which we then pass as function arguments. This is demonstrated in the code snippet below.

TUPLE AS A FUNCTION ARGUMENT

```
x   = [1, 2, 3]
tup = Tuple(x)

foo(tup) = sum(tup[1:2])

@code_warntype foo(tup)         # type stable
```

The approach presented should be your first option when transforming vectors to tuples. Nonetheless, there may be scenarios where defining the tuple inside the function is unavoidable. In such cases, there are a few alternatives.

Note first that simply passing the vector's number of elements as a function argument doesn't solve the issue. The reason is that the compiler generates method instances based on information about types, not values. This means that a function argument like `length(x)` merely informs the compiler

that the number of elements can be described as an object with type `Int64`, without providing any additional insight.

Instead, one effective solution is to define the tuple's length using a literal value, as demonstrated below.

NOT A SOLUTION

```
x   = [1, 2, 3]

function foo(x)
    tup = NTuple{length(x), eltype(x)}(x)

    sum(tup)
end

@code_warntype foo(x)        # type UNSTABLE
```

INFLEXIBLE SOLUTION

```
x   = [1, 2, 3]

function foo(x)
    tup = NTuple{3, eltype(x)}(x)

    sum(tup)
end

@code_warntype foo(tup)       # type stable
```

The downside of this solution is that it defeats the purpose of having generic code, as it restricts the function to tuples of a single predetermined size. To eliminate the type instability without constraining functionality, we need to introduce a more advanced solution. This is based on a technique known as **dispatch by value**. Since this approach is more complex to implement, *I recommend using it only when passing the tuple as a function argument is unfeasible*.

Next, we lay out the principles of dispatch by value, and then apply the technique to the specific case of tuples.

DEFINING DISPATCH BY VALUE

Dispatch by value enables passing information about values to the compiler. Nonetheless, implementing this feature requires a workaround, since the compiler only gathers information about types. The hack consists of creating a type that stores values as type parameters. In the case of tuples, this type parameter is simply the vector's number of elements.

The functionality is implemented via the built-in type `Val`, whose use is best explained through an example. Suppose a function `foo` and a value `a` that you wish the compiler to know. The technique requires defining `foo` with a type-annotated argument having no name, `::Val{a}`. After this, you must call `foo` passing an argument `Val(a)`, which instantiates a type with parameter `a`.

To illustrate the use of `Val`, we revisit an example included in previous sections. This considers a variable `y` that could be an `Int64` or `Float64`, contingent upon a condition. The ambiguity of `y`'s type is then transmitted to any subsequent operation, leading to type instability.

Dispatch by value is implemented by defining the condition as a type parameter of `Val`. In this way, the compiler will receive information about whether the condition is `true` or `false`, and therefore have knowledge about `y`'s type. This makes it possible to specialize its operations.

TYPE UNSTABLE

```julia
function foo(condition)
    y = condition ? 1 : 0.5      # either `Int64` or `Float64`

    [y * i for i in 1:100]
end

@code_warntype foo(true)         # type UNSTABLE
@code_warntype foo(false)        # type UNSTABLE
```

SOLUTION VIA "VAL"

```julia
function foo(::Val{condition}) where condition
    y = condition ? 1 : 0.5      # either `Int64` or `Float64`

    [y * i for i in 1:100]
end

@code_warntype foo(Val(true))    # type stable
@code_warntype foo(Val(false))   # type stable
```

> **Warning!**
> The function argument `Val` must be defined with `{}`, as types define their parameters with `{}`. Instead, `Val` must be called with `()`, as with any other function.

DISPATCHING BY VALUE WITH TUPLES

Let's now revisit the conversion of vectors to tuples. As we previously discussed, type instability arises because vectors don't store the size as part of their type information, leaving the compiler without sufficient information to determine the tuple's type.

Dispatch by value provides a solution to this issue: by passing the vector's length as a type parameter, the function call becomes type stable.

TYPE UNSTABLE

```
x = [1, 2, 3]

function foo(x, N)
    tuple_x = NTuple{N, eltype(x)}(x)

    2 .+ tuple_x
end

@code_warntype foo(x, length(x))        # type UNSTABLE
```

SOLUTION VIA "VAL"

```
x = [1, 2, 3]

function foo(x, ::Val{N}) where N
    tuple_x = NTuple{N, eltype(x)}(x)

    2 .+ tuple_x
end

@code_warntype foo(x, Val(length(x)))   # type stable
```

FOOTNOTES

[1.] Don't confuse `NTuple` as an abbreviation for the type `NamedTuples`. The "N" in the former case refers to a number "N" of elements.